

UNIVERSITY OF TORONTO, ON, Canada

WMO: 715080

Lat: 43.6658N

Lon: 79.3953W

Elev: 369

StdP: 14.50

Time Zone: -5.00 (NAE)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.8	7.0	-10.0	3.2	3.1	-3.7	4.6	9.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.8	87.5	72.7	84.6	71.2	82.1	69.7	75.5	83.7	73.8	81.3	72.2	79.0	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.0	124.4	80.1	71.4	117.3	78.5	69.8	111.0	77.1	39.2	84.3	37.7	81.4	36.2	79.1	84.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
N/A	N/A	N/A	DB	-4.3	92.4	6.1	3.6	-8.7	95.0	-12.3	97.1	-15.7	99.2	-20.2	101.8
			WB	-5.4	78.2	5.8	2.7	-9.5	80.2	-12.9	81.8	-16.1	83.3	-20.3	85.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.6	25.4	26.6	34.7	46.1	58.2	67.4	72.9	71.8	65.1	52.3	41.5	31.8
	DBStd	18.61	10.96	9.64	9.89	8.12	7.80	6.21	5.13	4.86	6.55	7.60	7.83	8.35
	HDD50	2973	763	655	480	166	14	0	0	0	0	62	269	564
	HDD65	6311	1227	1075	938	567	240	45	2	5	78	399	706	1029
	CDD50	2836	0	0	7	50	270	523	711	675	453	134	13	0
	CDD65	697	0	0	0	0	30	117	248	215	81	6	0	0
	CDH74	4264	0	0	0	4	188	710	1724	1229	391	18	0	0
	CDH80	1044	0	0	0	0	35	174	486	269	77	3	0	0
Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Precipitation	PrecAvg	33.3	2.4	2.2	2.6	2.7	2.8	2.7	2.9	3.2	3.0	2.5	3.1	3.0
	PrecMax	45.6	4.8	3.9	6.1	4.9	5.9	5.9	5.6	7.7	8.7	5.6	7.5	6.1
	PrecMin	25.5	0.8	0.8	0.5	1.2	1.0	0.7	0.8	0.7	0.4	0.5	0.5	0.9
	PrecStd	4.7	1.1	1.0	1.3	0.9	1.2	1.3	1.2	1.7	1.8	1.3	1.4	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.2	54.1	64.2	72.6	84.0	88.9	92.4	89.6	86.7	76.0	62.5	53.5
		MCWB	50.9	46.1	56.2	58.6	69.2	73.2	75.3	74.3	71.8	67.0	55.9	50.9
	2%	DB	47.7	46.8	57.0	66.8	79.2	84.8	88.2	85.8	81.9	70.8	58.2	48.8
		MCWB	45.4	42.5	50.5	54.9	65.9	70.5	73.1	73.0	70.5	63.3	52.9	45.9
	5%	DB	43.1	41.8	52.0	62.2	75.0	81.4	85.1	82.8	78.3	66.8	55.2	45.5
		MCWB	40.4	38.3	46.4	51.3	63.0	68.6	71.3	70.6	68.0	60.6	50.5	42.6
	10%	DB	39.5	38.6	47.9	58.4	70.8	78.1	82.3	80.2	75.3	63.6	52.5	42.5
		MCWB	37.1	35.6	42.7	49.2	59.9	66.8	69.8	69.1	67.0	58.3	48.5	39.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.1	48.0	58.0	61.3	72.0	75.6	78.7	77.8	74.6	69.2	57.2	51.1
		MCDB	52.8	50.8	62.0	68.5	80.5	85.6	88.6	85.2	82.6	74.6	61.0	52.9
	2%	WB	45.5	42.8	51.9	57.1	68.2	73.3	75.9	75.1	72.4	64.6	54.1	46.3
		MCDB	47.0	46.5	55.2	65.2	77.3	81.1	84.6	82.2	79.0	69.1	57.3	48.5
	5%	WB	40.7	38.6	47.0	53.5	64.8	70.9	74.2	73.2	70.5	61.8	51.2	43.2
		MCDB	42.5	41.3	51.3	60.3	72.3	78.5	81.8	80.1	76.6	65.4	54.4	45.3
	10%	WB	37.1	35.8	42.9	49.9	61.6	68.6	72.3	71.5	68.3	59.1	48.8	39.8
		MCDB	39.2	38.4	46.9	56.1	68.5	75.4	79.1	77.6	73.7	63.0	52.1	42.1
Mean Daily Temperature Range	5% DB	MDBR	11.2	11.2	12.0	13.4	14.8	14.7	14.8	14.0	13.6	11.5	10.1	9.3
		MCDBR	12.7	14.6	16.9	20.0	20.0	19.0	17.9	16.9	17.4	15.7	14.1	11.5
		MCWBR	12.2	12.6	12.7	13.4	11.3	10.3	8.8	8.4	9.1	10.8	11.6	10.9
	5% WB	MCDBR	12.3	14.1	16.5	19.1	18.2	17.3	15.8	15.0	15.1	14.3	13.0	11.5
		MCWBR	12.3	12.8	13.5	14.7	11.5	10.7	8.9	8.5	9.2	10.8	11.6	11.8
Clear-Sky Solar Irradiance	taub	0.301	0.321	0.334	0.382	0.412	0.440	0.434	0.420	0.387	0.366	0.321	0.302	
	taud	2.400	2.339	2.399	2.302	2.270	2.220	2.262	2.289	2.359	2.394	2.483	2.440	
	Ebn at Noon	262	276	287	280	273	265	265	265	265	256	254	251	
	Edh at Noon	24	30	33	39	42	44	42	39	34	29	23	22	
All-Sky Solar Radiation	RadAvg	447	682	1091	1406	1707	1853	1861	1635	1309	785	491	358	
	RadStd	42	62	109	149	181	135	137	83	120	71	62	38	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	-2.38	-2.81	N/A	N/A	N/A
(47) Regional (53 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page